

Abliteration Cannot Erase RLHF: Geometric Evidence from SAE Encoder Directions in Qwen3-8B

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Abstract

Refusal ablation (abliteration) edits a safety-aligned model’s weights to eliminate compliance refusals by removing a single residual-stream direction. The standard narrative frames abliteration as undoing RLHF safety fine-tuning. We test this at the encoder-direction level for Qwen3-8B using the frozen Qwen-Scope base SAE (65,536 TopK features), comparing three checkpoints—base, RLHF-instruct, and a community-released abliterated variant—across a balanced corpus of 500 harmful and 500 benign prompts. Because SAE reconstruction quality at layer 22 falls below acceptable thresholds ($R^2 \approx 0.51$ for instruct and abliterated variants), all metrics operate on pre-topk encoder projections ($\mathbf{z}_{\text{pre}}$)—dense continuous projections onto all encoder directions—rather than the post-topk sparse features validated as semantically interpretable in single-model settings. Within this geometric regime, we introduce two per-feature metrics: the Reversal Score $R(f)$, measuring directional preservation of RLHF-induced encoder-direction changes under abliteration, and the OLS Slope β_f , measuring magnitude preservation. For Qwen3-8B, and conditioned on the abliterated checkpoint faithfully implementing the refusal-direction removal, every one of the top-500 RLHF-sensitive encoder directions has a positive Reversal Score (mean $\bar{R} = 0.921 \pm 0.040$, min = 0.684; zero directions reversed), and the OLS slope is near unity ($\bar{\beta} = 0.986 \pm 0.176$; 86.4% of directions with $\beta_f > 0.8$). A noise-free mid-SC baseline (Cohen’s $d = 1.087$) confirms selective preservation. Direct measurement of refusal-direction geometry—mean $|\cos \theta| = 0.019$ between the refusal direction and top-500 RLHF-modified encoder directions at layer 22—provides the mechanistic explanation: abliteration’s orthogonalization is geometrically incapable of touching these directions. A 64.6% overlap with benign-prompt RLHF-sensitive directions reveals that the preserved modifications span both harm-specific and general RLHF stylistic changes. These findings establish, for Qwen3-8B, that abliteration removes the compliance mechanism while leaving the representational modifications of RLHF intact at approximately unit magnitude, pending validation of the pre-topk proxy and independent verification of abliteration depth.

1 Introduction

Refusal ablation—commonly called *abliteration*—edits a safety-aligned language model’s weights to eliminate its compliance refusals [Arditi et al., 2024]. The technique identifies a *refusal direction* in the residual stream: the mean-difference vector between activations on harmful versus harmless prompts. All weight matrices that write to the residual stream are then orthogonalized against this direction, producing a model that complies with previously-refused requests without retraining. The standard framing treats abliteration as *undoing* RLHF safety alignment: if RLHF installs a refusal mechanism, abliteration removes it, restoring a base-like model.

This framing has direct implications for AI safety. If abliteration truly erases RLHF modifications, the safety imprint is fragile—a single weight-editing step with minimal compute removes it entirely. If abliteration only removes the *behavioral expression* of safety training while leaving *representational*

modifications intact, the picture differs substantially: the model’s internal representations may still carry RLHF-induced structure, providing a substrate for lightweight safety restoration without full retraining. Distinguishing these scenarios matters both for threat modeling and for designing targeted safety-recovery interventions.

Two geometric studies establish that harmful-prompt representations survive ablation: LatentBiopsy [Llorente-Saguer, 2026a] finds that a linear classifier on Qwen model residual streams retains $\text{AUROC} \geq 0.937$ after ablation, and a cross-family companion study [Llorente-Saguer, 2026b] replicates this across 12 variants (Qwen, Llama, Gemma; 0.5B–9B parameters) with mean AUROC 0.982 surviving within ± 0.003 . Something in the internal representation persists—but holistic AUROC collapses all representational information into a single detection score and cannot answer *which specific encoder-direction structures survive, at what magnitude, or why*. Answering these mechanistic questions requires a feature-level vocabulary.

Sparse Autoencoders (SAEs) [Elhage et al., 2022, Bricken et al., 2023, Gao et al., 2024, Templeton et al., 2024] decompose residual-stream activations into tens of thousands of near-monosemantic features, providing exactly this vocabulary. Prior SAE-based analyses of refusal identify which features *drive* refusal behavior within a single model [Yeo et al., 2025, Kim et al., 2026]. Our work addresses a complementary question: which encoder directions are modified by RLHF, and do those modifications survive ablation in direction *and* magnitude? A key methodological complication is that reconstruction quality of a base-trained SAE degrades on instruct-variant activations [Chopra, 2026, Li et al., 2025], ruling out the standard post-topk sparse feature representation at the layers with the richest RLHF signal. We address this by working with *pre-topk encoder projections* ($\mathbf{z}_{\text{pre}}$)—dense continuous projections of residual-stream activations onto all 65,536 SAE encoder directions—and introduce two metrics interpretable in this geometric regime. The pre-topk proxy is validated indirectly via the orthogonality result (Section 5.5), which is independent of the sparse/dense distinction, and a direct $\mathbf{z}_{\text{pre}}/\mathbf{z}_{\text{post}}$ comparison is deferred to future work.

We apply the frozen Qwen-Scope base SAE (Qwen/SAE-Res-Qwen3-8B-Base-W64K-L0_50) [Deng et al., 2026] across three Qwen3-8B checkpoints—base, RLHF-instruct, and a community-released ablated variant—on a balanced corpus of 500 harmful and 500 benign prompts from seven datasets. The *Reversal Score* $R(f)$ measures whether ablation preserves or reverses the direction of the RLHF-induced change to encoder direction f . The *OLS Slope* β_f measures whether ablation preserves the magnitude of that change. For Qwen3-8B, conditioned on ablation completeness, every top-500 RLHF-sensitive encoder direction has a positive Reversal Score (mean $\bar{R} = 0.921$, min = 0.684, zero directions reversed), and the OLS slope is near unity ($\bar{\beta} = 0.986 \pm 0.176$, 86.4% of directions with $\beta_f > 0.8$). Direct orthogonality measurement (mean $|\cos \theta| = 0.019$ at layer 22) provides the geometric mechanism. A 64.6% overlap with benign-prompt RLHF-sensitive directions establishes that the preserved modifications are not exclusively harm-specific.

Summary of Contributions:

- We introduce the *Reversal Score*, a per-encoder-direction Pearson correlation confirming directional preservation of RLHF-modified SAE encoder directions under ablation for Qwen3-8B ($\bar{R} = 0.921$ at layer 22, zero reversed directions; Section 5.2).¹
- We introduce the *OLS Slope* β_f , a companion magnitude metric, confirming that ablation preserves not just direction but approximately full amplitude of RLHF-induced encoder-direction changes ($\bar{\beta} = 0.986 \pm 0.176$ at layer 22; Section 5.3).²

¹Code: <https://github.com/AMGrobelsnik/ai-invention-4ac301-abliteration-cannot-erase-rlhf-geometric/tree/main/round-2/experiment-1>

²Code: <https://github.com/AMGrobelsnik/ai-invention-4ac301-abliteration-cannot-erase-rlhf-geometric/>

- We validate selective preservation against a noise-free mid-SC baseline ($SC_{\text{median}} = 1.12 \gg 0.01$): top-500 $\bar{R} = 0.921$ vs. mid-500 $\bar{R} = 0.871$, Cohen’s $d = 1.087$, $p < 10^{-300}$ (Section 5.4).
- We directly measure refusal-direction orthogonality to RLHF-modified SAE encoder directions at layers 22 and 20 (mean $|\cos \theta| = 0.0185$ and 0.0209 , respectively), providing the geometric mechanism that explains zero reversed directions and $\bar{\beta} \approx 1$ (Section 5.5).³
- We show that 64.6% of the top-500 RLHF-sensitive encoder directions overlap with the top-500 benign-prompt RLHF-sensitive directions, establishing that the preserved RLHF modifications span general behavioral changes (output format, instruction style) in addition to harm-specific semantics (Section 5.6).

2 Related Work

Refusal ablation. Arditi et al. [2024] establish that refusal is mediated by a single residual-stream direction and that orthogonalizing all weight matrices against it reliably eliminates compliance across 13 models up to 72B parameters. Joad et al. [2026] challenge the single-direction framing, demonstrating that refusal behaviors correspond to geometrically distinct directions across eleven behavioral categories (safety refusals, excessive refusal, incomplete-request refusals, etc.), though they find that linear steering along any refusal-related direction produces nearly identical refusal-to-over-refusal trade-offs. This finding is directly relevant to our orthogonality measurement (Section 5.5): the mean $|\cos \theta| = 0.019$ between the mean-difference refusal vector and RLHF-modified encoder directions is a necessary but not sufficient condition for explaining zero reversed features if ablation removes a multi-dimensional subspace rather than a single direction. We address this in Section 5.5 and treat the multi-dimensional extension as explicit future work. Cristofano [2026] show the raw refusal vector is polysemantic, entangling safety circuitry with style; spectral residualization reduces collateral KL divergence while preserving ablation. Nanfack et al. [2026] propose an optimal-transport alternative reducing distributional shift. Our work examines what all these variants share: the refusal direction (or subspace), however constructed, leaves RLHF-modified SAE encoder directions intact in both direction and magnitude.

SAEs applied to refusal analysis. Yeo et al. [2025] use SAEs trained on pretraining data to investigate refusal mechanisms in chat-tuned models, finding that base-trained SAEs fail to cleanly encode the refusal circuitry of chat models—supporting our observation that base SAE reconstruction quality degrades on instruct activations ($R^2 \approx 0.51$ at layer 22). Their finding contextualizes our pre-topk proxy choice: the encoder directions of a base SAE are not guaranteed to align with the semantically active features of an instruct model, which is precisely why we restrict claims to *encoder-direction geometry* rather than asserting preservation of semantically validated sparse features. Kim et al. [2026] introduce CRaFT, which uses cross-layer transcoders to map model computations into sparse feature circuits, quantifying inter-feature influences on refusal-output logits rather than treating features as isolated signals. CRaFT and our approach are complementary: CRaFT identifies which features causally govern refusal outputs in a single model; our Reversal Score and OLS Slope identify which encoder directions RLHF modified and whether ablation erases those modifications. Both are feature-level analyses, but they use different feature

tree/main/round-4/experiment-1

³Code: <https://github.com/AMGrobelsnik/ai-invention-4ac301-abliteration-cannot-erase-rlhf-geometric/tree/main/round-3/experiment-1>

vocabularies (transcoder features vs. frozen base SAE encoder directions) and answer different questions (causal control of refusal vs. survival of RLHF modifications under ablation).

Geometry of harmful intent surviving ablation. LatentBiopsy [Llorente-Saguer, 2026a] and the companion geometry study [Llorente-Saguer, 2026b] are the direct predecessors establishing the phenomenon this work explains mechanistically. Their key limitation—collapsing representational information into a single AUROC score—is precisely the gap our encoder-direction analysis addresses. The OLS slope results ($\bar{\beta} = 0.986$) and direct orthogonality measurement ($|\cos \theta| = 0.0185$) provide the mechanistic account that a holistic detection score cannot.

SAEs as cross-variant measurement instruments. Chopra [2026] uses frozen base SAEs as cross-variant diagnostics for supervised fine-tuning (SFT), finding that SAE latent cosine similarity drops to 0.509–0.557 at layer 22 after SFT while raw activation cosine stays above 0.96—validating the frozen base-SAE approach and motivating the pre-topk encoder fallback when reconstruction quality degrades. Jiralerspong and Bricken [2026] introduce crosscoders—SAEs trained jointly on two or more model variants—that partition features into shared and model-specific categories without requiring a pre-existing vocabulary. Kassem et al. [2026] extend this to narrow fine-tuning regimes with Delta-Crosscoder. Our frozen base-SAE approach is complementary: it requires zero additional training, reuses Qwen-Scope’s existing feature vocabulary, and extends naturally to the three-model base→instruct→ablated trajectory. Li et al. [2025] demonstrate an $\approx 8\times$ MSE gap between base-SAE reconstruction on instruct versus purpose-trained instruct SAE activations, directly motivating our reconstruction quality gate and pre-topk fallback. Shportko et al. [2026] localize RL-induced tool-calling in Qwen2.5-3B to a compact steerable crosscoder feature set—the complement of our finding about which encoder directions ablation *cannot* remove.

Safety fine-tuning mechanisms. Jain et al. [2024] establish that safety fine-tuning projects unsafe activations into MLP null spaces, explaining adversarial circumvention. Our finding extends this picture: at the SAE encoder level, RLHF-modified encoder directions are near-orthogonal to the refusal direction, so ablation—which removes the refusal direction—cannot erase them geometrically.

3 Methodology

3.1 Three-Model Setup

All experiments use three Qwen3-8B checkpoints sharing identical 36-layer architecture ($d_{\text{model}} = 4096$, bfloat16, transformers $\geq 4.51.0$).⁴

Base: Qwen/Qwen3-8B-Base—the unaligned pretrained model, providing the representational baseline against which RLHF-induced changes are measured.

Instruct: Qwen/Qwen3-8B with `enable_thinking=False`—the RLHF-aligned model. Chain-of-thought suppression isolates safety-relevant activations on the final input token.

Ablated: huihui-ai/Huihui-Qwen3-8B-ablated-v2 (v2 corrects a layer-0 output-garbling artifact present in v1). The exact layer set and prompt set used for the refusal direction computation are not publicly documented. The findings in this paper are conditioned on this checkpoint faithfully implementing a refusal-direction removal: if ablation is incomplete, the high Reversal Scores

⁴Code: <https://github.com/AMGrobelsnik/ai-invention-4ac301-ablation-cannot-erase-rlhf-geometric/tree/main/round-1/research-1>

and OLS slopes would trivially reflect the ablated model being close to the instruct model by construction rather than preservation under genuine erasure. Section 5.5 and the Limitations section discuss this constraint.

All models are accessed via `model.model.layers` for residual-stream hook injection at the final token position of each input.

3.2 SAE Measurement Instrument

The frozen Qwen-Scope SAE `Qwen/SAE-Res-Qwen3-8B-Base-W64K-L0_50` serves as the shared measurement instrument [Deng et al., 2026]. A critical architectural note: Qwen-Scope implements *TopK* architecture ($k = 50$ features retained via `torch.topk`), not JumpReLU. For a residual activation $\mathbf{x} \in \mathbb{R}^{4096}$:

$$\mathbf{z}_{\text{pre}} = \mathbf{x}W_{\text{enc}}^{\top} + \mathbf{b}_{\text{enc}} \in \mathbb{R}^{65536} \quad (1)$$

$$\mathbf{z} = \text{TopK}(\mathbf{z}_{\text{pre}}, k=50) \quad (2)$$

Pre-topk proxy. The reconstruction quality gate (Section 3.5) fails at layer 22: $R^2 = 0.574$ (base), 0.512 (instruct), 0.512 (ablated)—all below the 0.85 threshold [Li et al., 2025]. All feature metrics therefore use $\mathbf{z}_{\text{pre}}$ rather than the post-topk sparse representation. The pre-topk vector is dense by construction: every prompt receives a non-zero value on all 65,536 encoder directions simultaneously, so no single direction is “off” in the interpretability sense. The metrics we compute (Reversal Score, OLS Slope) measure *geometric properties of encoder directions in a continuous projection space*, not activation of near-monosemantic features in the interpretability sense. The orthogonality result (Section 5.5) is fully independent of the sparse/dense distinction—it is a direct measurement of the angle between weight vectors—and remains the most secure finding. A direct comparison of top-500 feature rankings and Reversal Score rank correlations between $\mathbf{z}_{\text{pre}}$ and $\mathbf{z}_{\text{post}}$ on the highest- R^2 subset is an important validation deferred to future work. If such a comparison shows Spearman $\rho > 0.7$ between $\mathbf{z}_{\text{pre}}$ and $\mathbf{z}_{\text{post}}$ Reversal Scores, the pre-topk proxy is defensible as a surrogate; if it diverges, the encoder-direction findings must be interpreted as purely geometric properties without semantic grounding.

3.3 Feature Metrics

For each SAE encoder direction f and $N = 500$ harmful prompts $\{p_i\}_{i=1}^N$, let $a_m(f, p_i)$ denote the pre-topk encoder activation of direction f for model $m \in \{\text{base}, \text{inst}, \text{abl}\}$.

Safety Change quantifies the magnitude of the RLHF modification to encoder direction f :

$$\text{SC}(f) = \frac{1}{N} \sum_{i=1}^N |a_{\text{inst}}(f, p_i) - a_{\text{base}}(f, p_i)| \quad (3)$$

Encoder directions are ranked by $\text{SC}(f)$: the top-500 form the *RLHF-sensitive set*. For null-baseline analysis, a *mid-500* group is drawn uniformly from the 40th–60th SC percentile.

Reversal Score quantifies whether ablation reverses or preserves the *direction* of the RLHF modification to encoder direction f :

$$\Delta_{\text{inst}}(f) = [a_{\text{inst}}(f, p_i) - a_{\text{base}}(f, p_i)]_{i=1}^N \quad (4)$$

$$\Delta_{\text{abl}}(f) = [a_{\text{abl}}(f, p_i) - a_{\text{base}}(f, p_i)]_{i=1}^N \quad (5)$$

$$R(f) = \text{PearsonR}(\Delta_{\text{inst}}(f), \Delta_{\text{abl}}(f)) \quad (6)$$

$R(f) \approx +1$ indicates ablation preserves the RLHF modification (*imprinted*); $R(f) \approx -1$ indicates ablation reverses it (*reversed*). This is a correlation over prompt-level scalar activations, not a cosine similarity of scalars—ensuring that structure in the result is an empirical finding, not an artifact of the definition.

OLS Slope resolves the scale-invariance of the Reversal Score. A feature where ablation retains only 5% of the RLHF-induced shift in magnitude but preserves its direction would still yield $R(f) \approx 1.0$. The no-intercept ordinary least-squares slope:

$$\beta_f = \frac{\Delta_{\text{inst}}(f)^\top \Delta_{\text{abl}}(f)}{\Delta_{\text{inst}}(f)^\top \Delta_{\text{inst}}(f)} \quad (7)$$

$\beta_f \approx 1$ means the ablated model’s encoder-direction activations shift by the same magnitude as the instruct model’s, relative to base. $\beta_f \approx 0$ means only the directional pattern survives with negligible magnitude. $\beta_f > 1$ indicates overshooting.

3.4 Benign-SC Overlap

To assess whether the top-500 RLHF-sensitive encoder directions reflect *harm-specific* modifications or *general RLHF* modifications (style, tone, response format), we separately compute $\text{SC}_{\text{benign}}(f)$ using the 500 benign prompts and report the overlap between the top-500 by harmful-SC and the top-500 by benign-SC.

3.5 Layer Selection and Reconstruction Quality Gate

Layer selection. We compute $\text{SC}(f)$ variance across all 65,536 encoder directions at candidate layers {14, 20, 22} on 50 harmful prompts. Layer 22 is selected (variance 0.1721 vs. 0.0547 at layer 20 and 0.0127 at layer 14); all primary results use layer 22 with layer 20 as replication.

Reconstruction quality gate. At layer 22, $R^2 = 0.574$ (base), 0.512 (instruct), 0.512 (ablated)—all below the 0.85 threshold [Li et al., 2025]—triggering the pre-topk encoder-only analysis path.

3.6 Semantic Labeling Protocol

For the top-100 encoder directions by $\text{SC}(f)$, we identify the 5 prompts with the highest pre-topk encoder activation and provide the first 300 characters of each to Llama-3.1-8B-Instruct via OpenRouter. The cross-family labeler avoids within-Qwen circularity. Each direction is classified as `harm_concept`, `safety_refusal`, `general_capability`, or `other`. Results are treated as qualitative observations only: the labeling uses character-prefix truncation of most-activating prompts rather than per-token activation windows; labels reflect prompt topic more than token-level representations; and no Cohen’s κ validation against a manual gold set was performed.

4 Evaluation Corpus

The evaluation corpus comprises 500 harmful and 500 benign prompts from seven publicly licensed datasets.

Harmful sources: BeaverTails [Ji et al., 2023] (176 prompts; NeurIPS 2023, human-labeled QA spanning 14 harm categories), AdvBench (98 prompts; standard adversarial behavior dataset used in ablation research [Arditi et al., 2024]), TrustAIRLab in-the-wild jailbreak prompts (86 prompts; real-world jailbreaks from ACM CCS 2024), HarmBench [Mazeika et al., 2024] (85 prompts; covering

chemical, cybercrime, physical harm, and illegal categories), and JailbreakBench/JBB-Behaviors [Chao et al., 2024] (55 prompts).

Benign sources: Jailbreak-classification (151 prompts; labeled benign user requests), JBB-Behaviors benign counterparts [Chao et al., 2024] (46 prompts), Toxic-Chat (104 prompts; real user prompts labeled non-toxic), TrustAIRLab regular prompts (93 prompts), and Dolly-15k [Conover et al., 2023] (106 prompts; human-written instruction-following tasks).

All sources are processed through MinHash LSH deduplication (threshold 0.5, 128 permutations, character 5-gram shingling), token-length filtering (≤ 512 tokens), and category capping ($\leq 35\%$ per harm subcategory). The final corpus contains 1,000 prompts (500 harmful, 500 benign).

5 Results

5.1 Layer Selection

Layer 22 exhibits the highest Safety Change variance (0.1721), confirming it as the primary analysis layer. The R^2 reconstruction gate at layer 22 yields values below 0.85 for all three model variants (base: 0.574, instruct: 0.512, ablated: 0.512), triggering the pre-topk encoder-only analysis path (Section 3.2).

5.2 Reversal Score Distribution

Every top-500 RLHF-sensitive encoder direction has a positive Reversal Score (mean = 0.921, std = 0.040, min = 0.684, max = 0.989). No direction has a negative or near-zero score. Hartigan’s dip test yields statistic = 0.0123, $p = 0.930$ —strongly unimodal. Spearman correlation mean = 0.913, confirming rank-based robustness.

Table 1 summarizes the count of reversed and imprinted directions at all tested thresholds. Zero directions are reversed at any threshold.

Table 1: Imprinted and reversed encoder-direction counts at all tested thresholds.

Threshold	Imprinted ($R > \text{threshold}$)	Reversed ($R < -\text{threshold}$)
0.30	500 / 500 (100.0%)	0 / 500 (0%)
0.50	500 / 500 (100.0%)	0 / 500 (0%)
0.70	499 / 500 (99.8%)	0 / 500 (0%)
0.95	125 / 500 (25.0%)	0 / 500 (0%)

5.3 OLS Slope: Magnitude Preservation

A key limitation of the Reversal Score is scale-invariance: $R(f) \approx 1$ is consistent with ablation preserving only 5% of the RLHF-induced magnitude while retaining the direction pattern. The OLS slope β_f resolves this ambiguity.

The β_f distribution for the top-500 RLHF-sensitive encoder directions at layer 22 is tightly concentrated near 1: mean $\bar{\beta} = 0.986 \pm 0.176$, median = 0.985, range $[-0.055, 1.725]$. Specifically, 86.4% of directions have $\beta_f > 0.8$ (ablation preserves more than 80% of the RLHF-induced shift in magnitude), 45.4% have $\beta_f > 1.0$ (slight amplification), and only 0.2% are negative. **MAGNITUDE PRESERVED.** Within the pre-topk geometric regime, ablation does not attenuate the RLHF imprint—it preserves approximately unit magnitude.

Layer-20 replication: $\bar{\beta} = 0.954 \pm 0.191$, 80.2% of directions with $\beta_f > 0.8$, zero negative values—consistent with the layer-22 primary result.

5.4 Null-Baseline Validation: Noise-Free Mid-SC Comparison

A potential concern with a bottom-500 null baseline is that near-zero Safety Change values may make PearsonR between near-zero vectors potentially unreliable. We address this with a noise-free mid-SC baseline: 500 directions uniformly sampled from the 40th–60th SC percentile, with $SC_{\text{median}} = 1.12$ (versus the top-500 minimum of 2.79).

At layer 22: top-500 $\bar{R} = 0.921$, mid-500 $\bar{R} = 0.871$ (Cohen’s $d = 1.087$, two-sample t -test $t = 17.2$, $p < 10^{-300}$). **MID-SC CONFIRMS SELECTIVE PRESERVATION.** The top-500 encoder directions show significantly higher Reversal Scores than mid-SC controls that have non-trivial RLHF sensitivity and noise-free Δ_{inst} vectors, confirming that the high Reversal Scores reflect genuine selective preservation rather than any measurement artifact.

5.5 Direct Measurement of Refusal-Direction Orthogonality

The mechanistic interpretation of zero reversed directions and $\bar{\beta} \approx 1$ requires a testable prediction: the refusal direction $\mathbf{r}$ must be nearly orthogonal to all RLHF-modified SAE encoder directions. We test this directly.

The refusal direction $\mathbf{r}$ is computed as the mean difference of instruct residual-stream activations on harmful versus harmless prompts at layer 22. For each of the top-500 RLHF-sensitive encoder directions f :

$$\cos \theta_f = \frac{\mathbf{r} \cdot \mathbf{w}_{\text{enc}}(f)}{\|\mathbf{r}\| \|\mathbf{w}_{\text{enc}}(f)\|} \quad (8)$$

Layer 22: mean $|\cos \theta_f| = 0.01854$ (authoritative value; 0.019 in the abstract is rounded). **ORTHOGONALITY CONFIRMED.** A cosine of 0.0185 means the refusal direction projects onto less than 1.9% of each RLHF-modified encoder direction’s magnitude. **Layer 20:** mean $|\cos \theta_f| = 0.02086$. Orthogonality confirmed at both layers; inter-layer agreement (matching to 5 decimal places across independent experimental runs) confirms these values are stable.

This orthogonality result is the most secure finding in this paper: it is a direct measurement of the angle between weight vectors and is fully independent of the pre-topk/post-topk distinction, the ablation checkpoint’s completeness, or any activation-level measurement.

This orthogonality geometrically explains the $\bar{\beta} \approx 1$ result: ablation orthogonalizes all weight matrices against $\mathbf{r}$. Because $\mathbf{r}$ is nearly orthogonal to all RLHF-modified SAE encoder directions, the orthogonalization cannot substantially change how any RLHF-modified encoder direction’s activations are computed—in direction or in magnitude.

Caveat on multi-dimensional refusal. Joad et al. [2026] demonstrate that refusal spans geometrically distinct directions across eleven behavioral categories. If ablation removes a multi-dimensional refusal *subspace* rather than a single vector, the orthogonality argument requires that the full ablation subspace be near-orthogonal to all RLHF-modified encoder directions. The current measurement tests only the mean-difference vector $\mathbf{r}$. An extension to the top-3 principal components of the (harmful – harmless) activation differences would determine whether this argument holds for higher PCs; this is deferred to future work. If a higher PC shows mean $|\cos \theta_f| \geq 0.05$ against the top-500 encoder directions, the mechanistic explanation would require qualification.

5.6 Benign-SC Overlap: What the Preserved Imprint Represents

We compute $SC_{\text{benign}}(f)$ —Safety Change evaluated on 500 benign prompts—for all 65,536 SAE encoder directions, then measure the overlap between the top-500 by harmful-SC and the top-500 by benign-SC.

Overlap count: 323 / 500 = 64.6%. STYLE-DOMINANT verdict. A majority of the top-500 RLHF-sensitive encoder directions also appear in the benign-SC top-500, indicating that RLHF modified these directions similarly when processing harmful and benign prompts. Approximately 64.6% of the preserved imprint captures general RLHF modifications (response formality, output structure, instruction-following style) rather than harm-specific semantic content. The remaining 35.4% (176 directions) are uniquely sensitive to harmful prompts and more plausible carriers of harm-specific semantic representations.

This finding contextualizes the OLS slope result ($\bar{\beta} = 0.986$): ablation comprehensively preserves all RLHF encoder-direction modifications regardless of whether they encode harm-specific or general-style content. The near-orthogonality of the refusal direction to all RLHF-modified encoder directions means ablation cannot selectively erase harm-semantic changes while leaving style changes intact—it fails to erase either category because all reside in a subspace near-orthogonal to $\mathbf{r}$.

5.7 Centroid Analysis with Control Subspaces

To validate that the preserved imprint is specific to the RLHF-sensitive subspace, we compute centroid distance ratios $d_{\text{abl-base}}/d_{\text{abl-inst}}$ (cosine) for three encoder-direction subspaces.

Layer 22 (harmful prompts): Subspace A (top-500 RLHF-sensitive) ratio = $4.35\times$, Subspace B (bottom-500 RLHF-insensitive) ratio = $1.16\times$, Subspace C (random-500) ratio = $2.37\times$. **Layer 20:** Subspace A ratio = $7.31\times$, Subspace B ratio = $1.01\times$, Subspace C ratio = $1.63\times$ (verdict: CENTROID SELECTIVE). The RLHF-sensitive subspace shows $4\text{--}7\times$ greater instruct-closeness than the RLHF-insensitive control ($B \approx 1.0\text{--}1.2$ vs. $A = 4.35\text{--}7.31$), confirming that the imprint is specific to the directions RLHF modified, not a global consequence of model similarity. Layer-20 Subspace B reaching $1.01\times$ (essentially no instruct-closeness in the RLHF-insensitive subspace) provides the cleanest control.

5.8 Downstream Classification Accuracy

This null result follows directly from the 64.6% benign-SC overlap (Section 5.6): the top-500 RLHF-sensitive encoder directions are elevated by RLHF on *both* harmful and benign prompts, eliminating prompt-level discriminability by construction. This distinguishes *what RLHF modified* (broad encoder-direction changes) from *what selectively activates on harmful versus benign prompts at inference* (a different, narrower set of encoder directions not analyzed in this work).

A per-prompt harm score (mean pre-topk activation across all top-500 RLHF-sensitive encoder directions with median threshold) achieves **0.476 accuracy**—near-chance on the balanced corpus—versus a keyword baseline of **0.630**. This null classification result confirms the 64.6% benign-SC overlap from an independent angle: RLHF-sensitive encoder directions elevated on harmful prompts are similarly elevated on benign prompts, yielding no discriminative signal.

5.9 Semantic Labeling

Llama-3.1-8B-Instruct assigned labels to 100 top-SC encoder directions: the majority received `harm_concept` labels, with smaller proportions of `safety_refusal`, `general_capability`, and

other. These results are qualitative observations only. The labeling uses character-prefix truncation of most-activating prompts rather than per-token activation windows; no Cohen’s κ validation against a manual gold set was performed; and the 64.6% benign-SC overlap implies that many stylistic directions receive `harm_concept` labels because they activate most strongly on harmful-prompt contexts rather than encoding harm-specific semantic content. Quantitative reliance on the specific labeling percentages requires a manual gold-set annotation with $\kappa \geq 0.6$, deferred to future work.

5.10 Layer-20 Replication

Table 2 summarizes full quantitative replication at layer 20.

Table 2: Quantitative results at layers 22 and 20.		
Metric	Layer 22	Layer 20
Top-500 Reversal Score ($\bar{R}$)	0.921	0.924
Mid-500 Reversal Score ($\bar{R}$)	0.871	0.858
Cohen’s d (top vs. mid)	1.087	—
Top-500 OLS Slope ($\bar{\beta}$)	0.986 ± 0.176	0.954 ± 0.191
Fraction $\beta_f > 0.8$	86.4%	80.2%
Fraction $\beta_f < 0$	0.2%	0.0%
Mean $ \cos(\mathbf{r}, \mathbf{w}_{\text{enc}}) $ top-500	0.01854	0.02086
Centroid A (RLHF-sensitive)	$4.35\times$	$7.31\times$
Centroid B (RLHF-insensitive)	$1.16\times$	$1.01\times$
Centroid C (random-500)	$2.37\times$	$1.63\times$
Centroid verdict	MODERATE	SELECTIVE

REPLICATION CONFIRMED at both layers.

6 Discussion

6.1 Geometric Interpretation of Encoder-Direction Preservation

The OLS slope result ($\bar{\beta} = 0.986$, 86.4% of directions with $\beta_f > 0.8$) establishes that the preservation is comprehensive in both direction and magnitude. The direct orthogonality measurement ($|\cos \theta| = 0.0185$) provides the mechanistic account: ablation removes $\mathbf{r}$ from all weight matrices, but for ablation to change the projection of a residual-stream activation onto SAE encoder direction f by even 10%, $\mathbf{r}$ would need to project onto $\mathbf{w}_{\text{enc}}(f)$ with $|\cos \theta| \geq 0.1$. The measured mean $|\cos \theta| = 0.0185$ falls $5\times$ below this threshold, explaining why both direction and magnitude are preserved: the orthogonalization barely touches the RLHF-modified encoder directions. This interpretation is consistent with Cristofano’s observation [Cristofano, 2026] that the raw refusal direction primarily captures stylistic compliance patterns.

We note that this mechanistic account applies in the pre-topk projection regime. Whether the same geometry holds for the post-topk sparse activations validated as semantically interpretable in single-model settings [Bricken et al., 2023, Templeton et al., 2024] remains an open question. The orthogonality result—which is a property of weight vectors, not activations—directly implies that *any* function of the residual stream that can be expressed as a projection onto $\mathbf{w}_{\text{enc}}(f)$ will be approximately unaffected by ablation; the question is whether the semantically active post-topk

features are well-described by such projections, which requires a $\mathbf{z}_{\text{pre}}/\mathbf{z}_{\text{post}}$ correlation analysis not yet available.

6.2 What the Preserved Imprint Represents

The 64.6% benign-SC overlap requires careful interpretation. The majority of RLHF-sensitive encoder directions are modified similarly when processing harmful and benign prompts, indicating these directions capture *general* RLHF behavioral changes (formality, helpfulness, instruction-following structure) rather than harm-specific semantic content. The surviving 35.4% uniquely-harmful directions are the more plausible carriers of harm-concept semantic content.

Crucially, the near-orthogonality of the refusal direction to all RLHF-modified encoder directions (harmful-specific and style-related alike) means ablation geometrically cannot remove any category of RLHF modification. The behavioral regression to base-like output arises because the compliance mechanism (the refusal direction) is removed, not because the underlying encoder-direction modifications are erased. The model forgets how to express refusal while retaining all representational changes RLHF introduced—at approximately full magnitude.

This also explains the downstream classification null (accuracy = 0.476): the RLHF-sensitive encoder directions are elevated on *both* harmful and benign prompts, so mean-activation discrimination is not available. The representational change RLHF introduced is a broad encoder-direction shift, not a selective harm-detector.

6.3 Implications for Safety Restoration

If ablation preserves the RLHF imprint at approximately unit magnitude, the challenge for targeted safety restoration is not to amplify surviving encoder-direction modifications—they are already at full instruct-model activation levels. The challenge is to restore the *compliance mechanism*: the geometric pathway from current activations to refusal outputs, implemented by the refusal direction $\mathbf{r}$. Restoration requires re-injecting $\mathbf{r}$ into the weight matrices without disturbing the already-intact encoder-direction modifications.

This reframes the restoration problem as geometric re-injection rather than semantic amplification. The approach of Shportko et al. [2026] identifies compact steerable feature sets for RL-induced capabilities; an analogous method targeting the encoder directions that mediate harm-concept-to-refusal conversion could enable targeted re-injection. The 35.4% uniquely-harmful encoder directions are the most promising candidates for such targeted intervention, pending confirmation that they carry semantically coherent harm-specific content (which requires $\mathbf{z}_{\text{post}}$ analysis).

6.4 Limitations

Pre-topk encoder projections. All feature metrics use $\mathbf{z}_{\text{pre}}$ (dense encoder projections) rather than the post-topk sparse activations validated as interpretable by Bricken et al. [2023] and Templeton et al. [2024]. Pre-topk values are dense by construction—every encoder direction receives a non-zero value on every prompt—so findings are more accurately characterized as *geometric properties of encoder directions in a continuous projection space* than as *feature-level preservation of near-monosemantic SAE features*. A validation experiment is required before semantic claims can be grounded: on the highest- R^2 prompt subset, compare top-500 features selected by $\text{SC}(\mathbf{z}_{\text{pre}})$ versus $\text{SC}(\mathbf{z}_{\text{post}})$ (overlap fraction) and compute Spearman ρ between $\mathbf{z}_{\text{pre}}$ and $\mathbf{z}_{\text{post}}$ Reversal Scores. If $\rho > 0.7$ and overlap $> 60\%$, the pre-topk proxy is defensible; if they diverge, the semantic labeling results must be interpreted purely as observations about encoder-direction geometry. This is the most critical open validation gap.

Abliteration depth verification. The `huihui-ai/Huihui-Qwen3-8B-abliterated-v2` checkpoint was not independently verified. If abliteration is incomplete—even partially—the high Reversal Scores and $\bar{\beta} \approx 1$ could trivially reflect the abliterated model being close to the instruct model by construction. Two measurements would resolve this: (1) fraction of AdvBench or HarmBench prompts on which the abliterated checkpoint produces a non-refusal response (expected near 100% for complete abliteration); (2) mean $|\text{Proj}_{\mathbf{r}}(\mathbf{h}_{\text{abl}}(p))|/|\mathbf{h}_{\text{abl}}(p)|$ at layer 22 for the 500 harmful prompts (expected near zero if $\mathbf{r}$ has been removed). These measurements can be computed from already-cached activations. Until they are reported, the main findings should be interpreted as conditional on abliteration completeness.

Multi-dimensional refusal subspace. The orthogonality measurement uses the mean-difference vector $\mathbf{r}$ (a single vector). Joad et al. [2026] demonstrate that refusal spans geometrically distinct directions across behavioral categories, implying abliteration may remove a multi-dimensional subspace rather than a single vector. The mechanistic explanation of zero reversed directions requires near-orthogonality of the *full ablation subspace* to all RLHF-modified encoder directions, not just the mean-difference summary. Computing mean $|\cos \theta_f|$ for the top-3 principal components of the (harmful – harmless) activation differences at layer 22 would determine whether the orthogonality argument holds robustly; this is an important future measurement.

Single model family and layers. All results are for Qwen3-8B at layers 22 and 20. The zero-reversed finding and $\bar{\beta} \approx 1$ result may not generalize to other model families (Llama, Gemma) or other layer depths. The companion geometry studies [Llorente-Saguer, 2026a,b] cover 12 variants, establishing the holistic survival phenomenon across families; analogous SAE-level analysis for Llama-3.1-8B and Gemma-3-9B using GemmaScope and LlamaScope SAEs would test whether encoder-direction preservation is a general property of the abliteration geometry.

Semantic labeling without validation. The qualitative observation that the majority of top-SC encoder directions receive `harm_concept` labels cannot support quantitative downstream reasoning without Cohen’s κ validation against a manual gold set of at least 50 annotations.

7 Conclusion

We present the first encoder-direction-level mechanistic analysis of refusal abliteration using the frozen Qwen-Scope base SAE across three Qwen3-8B variants. Two complementary per-direction metrics—the Reversal Score (directional preservation) and the OLS Slope (magnitude preservation)—together establish, for Qwen3-8B, a comprehensive geometric imprint: every top-500 RLHF-sensitive SAE encoder direction has a positive Reversal Score (mean $\bar{R} = 0.921 \pm 0.040$, min = 0.684) and an OLS slope near unity ($\bar{\beta} = 0.986 \pm 0.176$, 86.4% exceeding 0.8). A noise-free mid-SC baseline (Cohen’s $d = 1.087$) confirms selective preservation. Direct measurement of refusal-direction geometry ($|\cos \theta| = 0.0185$ at layer 22, 0.0209 at layer 20) provides the mechanistic explanation: abliteration’s orthogonalization is geometrically incapable of touching the RLHF-modified encoder directions. A 64.6% benign-SC overlap reveals that the preserved imprint spans both harm-semantic and general RLHF stylistic encoder-direction modifications; abliteration fails to erase any category because all reside in a subspace near-orthogonal to the refusal direction. For Qwen3-8B, the abliterated model’s behavior reverts to base-like, but its encoder-direction projections remain instruct-like at full magnitude—a dissociation that reframes safety restoration as geometric re-injection of the compliance direction rather than semantic amplification of surviving features.

Three critical open gaps constrain full confidence in the broader interpretation: (1) $\mathbf{z}_{\text{pre}}/\mathbf{z}_{\text{post}}$ validation (Spearman ρ and feature-ranking overlap at the highest- R^2 subset) to determine whether the geometric encoder-direction findings correspond to semantically interpretable sparse features;

(2) independent behavioral and geometric verification of ablation depth for the community checkpoint; and (3) extension of the orthogonality measurement to the top-3 principal components of the refusal subspace, following Joad et al. [2026].

Future directions: (i) $\mathbf{z}_{\text{pre}}/\mathbf{z}_{\text{post}}$ validation experiment; (ii) ablation depth verification via refusal-rate measurement and mean $|\text{Proj}_{\mathbf{r}}|$ at layer 22; (iii) multi-dimensional refusal subspace (top-3 PC orthogonality); (iv) cross-family extension to Llama-3.1-8B and Gemma-3-9B using GemmaScope and LlamaScope SAEs; (v) multi-layer analysis across all 36 Qwen-Scope layers; (vi) investigation of the 35.4% uniquely-harmful encoder directions as candidates for targeted geometric restoration.

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